

Quiz 5A

Name: _____ **Roll No:** _____ **Section:** _____

Question 1: marks are dependent on correct answer of Question 2 (10 + 5 = 15 marks)

For the following code show the status of the stack and registers when label 'start:' is reached. Starting address of stack is **0200H** and starting address of memory is **0500H**.

```

1      .data
2          array word 3, -1, -4, 2
3          res word ?
4      .code
5      main Proc
6          mov esi, offset array
7          push esi
8          mov ecx, lengthof array
9          push ecx
10         mov esi, offset res
11         push esi
12         call sampleproc
13         mov ax, res
14     INVOKE ExitProcess,0
15     main ENDP
16
17     sampleproc proc
18         push ebp
19         mov ebp, esp
20         mov esi, [ebp+16]
21         mov ecx, [ebp+12]
22         mov edi, [ebp+8]
23         mov dx, 100
24     start:
25         cmp [esi], dx
26         jnl lower
27         jmp higher
28     lower:
29         mov dx, [esi]
30     higher:
31         add esi, 2
32         loop start
33         mov [edi], dx
34         pop ebp
35         ret 12
36     sampleproc endp
37     END main
38
39

```

[illegible]

Question 2: (5 marks)

Understand the above code (in question 1) and describe its operation in single line

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Question 3: (10+5+5 =20 marks)

Fill stack for your understanding and update register after every instruction. Stack address starts **0100H**

[illegible][illegible]